

Assignment /

Assignment #:20

Topic : String

Level 1: Easy

1. STRING OPERATION

Given a string and reverse the using Negative slicing.

Input format

The first line of each test case consists of string with Unicode character.

Output format

Print the reversed string.

Constraints

Using Negative slicing operation.

Sample test Cases:

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	"Mycomputer"	"Student"	"playing"	"videogame"
Output	"retupmocyM"	"tnedutS"	"gniypal"	"emagoediv"

2. STRING METHODS

Given an string and find the last position of a substring of given input in a program.

Input format

String with sequence of words

Output format

Print the substring of give input

Constraints

Using rfind() methods

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	str1 = "Emma is a data scientist who knows Python. Emma works at google."	str1 = "Python is machine language . python developer."	str1 = " data scientist who knows Python. Google hire them."	str1 = "He Knows you before. works at google."
Output	Last occurrence of Emma starts at index 43	Last occurrence of python starts at index 29	Last occurrence of Google starts at index 34	Last occurrence of works starts at index: 21

3. STRING BUILT IN METHODS

Given a string to replace each special symbol with # in the following string

Input format

The input string with special symbol.

Output format

Print replaced string

Constraints

Input should have special symbol.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
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Input	str1 = '/*Jon is @developer & musician!!'	str1 = ui!w@ghy^ui'	str1 = 'phy%chy\$bio@'	str1 = 'g@d&he\$tea*%!'
Output	##Jon is #devel musician##	ui#w#ghy#ui'	phy#chy#bio#	g#d#he#tea###

4. SEARCH ALPHANUMERIC STRING

Rohan have a string with sequence of words Can you help him to find the alphanumeric word of given string?

Input format

The string with sequence of words or sentences

Output format

Print the alphanumeric string.

Constraints

String contain alphanumeric character.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	"Emma25 is Data scientist50 and AI Expert"	"Hi21 this is good35"	"raja1 and his2 daughter"	"krishnan88 and play6 game"
Output	Emma25 scientist50	Hi21 good35	raja1 his2	krishnan88 play6

5. LOWERCASE LETTERS SHOULD COME FIRST

Gayathri is given string contains a combination of the lower- and upper-case letters. Can you help her arrange the characters of a string so that all lowercase letters should come first?

Input format

The input string contains lower- and upper-case letters

Output format

Print the lowercase letter first

Constraints

String with combination of the lower- and upper-case letters.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	str1 = PYnAtive	str1=KarTHik	str1=RoHAnkuMaR	str1=HeAdInG
Output	ntivPYAE	arikKTH	onuaRHAKMR	ednHAIG

Level 2: Medium

1. CONCATENATION OF TWO STRING TUPLES

Rahul is working with records. He has a problem in which we may need to perform String concatenation of tuples. This problem can occur in day-day programming. Can you complete the task for him?

Input format

The first line of each test case consists of tuple 1.

The second line of each test case consists of tuple 2.

Output format

Print CONCATENATED tuple

Constraints

Using concatenation Operator

Sample test Cases

Test Cases	Test Case1:	Test Case2:

Input	The original tuple 1: ('Manjeet', 'Nikhil', 'Akshat') The original tuple 2: (' Singh', ' Meherwal', ' Garg')	The original tuple 1: ('My', 'note', 'sun') The original tuple 2: (' World', ' book', ' set')
Output	The concatenated tuple: ('Manjeet Singh', 'Nikhil Meherwal', 'Akshat Garg')	The concatenated tuple: ('My World', 'Note book', 'sun set')

2. INDEXING

Individual characters of a String can be accessed by using the method of Indexing,so help rohan to Accessing characters in Python String by indexing technique

Input format

Input is declared in string1 by different test cases

Output format

print first character of string and last character of string

Constraints

Only Integers are allowed to be passed as an index, float or other types that will cause a TypeError.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	Initial String: Greek strings	Initial String: Native city	Initial String: Book cover	Initial String: Back bag weight
Output	First character of String G	First character of Str N	First character of Strin B	First character is:

	Last character of String s	Last character of Str y	Last character of Strin r	B Last character is: t
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3. STRING SLICING

Mohan wants to access a range of characters in the String, the method of slicing is used. Slicing in a String is done by using a Slicing operator (colon). So, help him for this task completion?

Input format

The input string1 is declared in program with different characters

Output format

Print sliced string of given string1.

Constraints

Using string slicing technique

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	string1="goodstrainsforpeople"	string1="nothing is difficult"	string1="everything is possible"	string1="gods with you"
Output	Slicing characters from 3-12: dstrainfo	Slicing characters from 3-12: hing is di	Slicing characters from 3-12: rything is	Slicing characters from 3-12: s with you

4. UPDATING FROM A STRING

Magizh found difficulties in updating of characters from a String is not allowed. This will cause an error because item assignment or item deletion from a String is not supported because strings are immutable, so help him for updation of string?

Input format

Getting a string2 input to update string1

Output format

Print updated string

Constraints

Using indexing technique

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	Initial String: Hello, I'm a student	Initial String: need a help	Initial String: Take this book	Initial String: success is not permanent
Output	Updating character at 2nd Index: Heplo, I'm a student Heplo, I'm a student	Updating character at 2nd Index: nepd a help nepd a help	Updating character at 2nd Index: Tape this book Tape this book	Updating character at 2nd Index: supcess is not permanent supcess is not permanent

5. Formatting of Strings

Strings in Python can be formatted with the use of format() method which is a very versatile and powerful tool for formatting Strings. Create program using this format() method.

Input format

Getting an input to format with particular order

Format defined in program

Output format

Print a formatted string

Constraints

Using format() method

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	string1="Good for life"	string1="Love Is God"	string1="Nothing Is Bad"	string1="Too Much Thing"
Output	Print String in default order: Good For Life Print String in Positional order: For Good Life Print String in order of Keywords: Life For Good	Print String in default order: Love Is God Print String in Positional order: Is Love God Print String in order of Keywords: God Is Love	Print String in default order: Nothing Is Bad Print String in Positional order: Is Nothing Bad Print String in order of Keywords: Bad is Nothing	Print String in default order: Too Much Thing Print String in Positional order: Much Too Thing Print String in order of Keywords: Thing Much Too

Level 3: Hard

1. COUNT STRING

Shalini is given a string and help her to Count and print out the number of lower case, upper case, and non-letters.

Input format

Getting an input string with different format

Output format

You need to print number of lower case, upper case, and non-letters.

Constraints

Contains combination of words

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	abcXYZ784*&^def	cirCULarCom@#s%^	REmoTE@#con&*	lapTOP12(&^)
Output	Lower case letters 6 Upper case letters 3 Non - letters: 6	Lower case letters 8 Upper case letters 4 Non - letters: 4	Lower case letters 5 Upper case letters 4 Non - letters: 4	Lower case letters : Upper case letters:3 Non - letters: 6

2. MIXED STRING

Rohan has two strings, s1 and s2. And to create a new string s3 made of the first char of s1, then the last char of s2, Next, the second char of s1 and second last char of s2, and so on. Any leftover chars go at the end of the result. Can you help him to complete the task?

Input format

The first string str1 is declared in program

The second string str2 is declared in program

Output format

Print the third string str3

Constraints

Input should be in string format

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	s1 = "Abc" s2 = "Xyz"	s1 = " Def" s2 = " Hji"	s1 = " Klo" s2 = " Pyt"	s1 = "Mno" s2 = "Kgf"
Output	AzbycX	Diejfh	KtlyoP	MFngoK

3. ESCAPE SEQUENCING IN PYTHON

Rohan printing Strings with single and double quotes in it causes SyntaxError because String already contains Single and Double Quotes and hence cannot be printed with the use of either of these. Hence, can you help Rohan to print such a String either Triple Quotes are used or Escape sequences are used to print such Strings?

Input format

The first line of Escape sequences starts with a backslash and can be interpreted differently.

The input string with single quotes is used to represent a string

Then all the single quotes present in the string must be escaped.

Output format

You need to print single quoted, double quoted and backslash string

Constraints

The single quotes present in the string must be escaped and same is done for Double Quotes.

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	String1 = ' I'm a "Geek" '	String1 = ' Escaped "Easy" '	String1 = ' It's a "Good" '	String1 = ' = "Concept" '
Output	Initial String with use of Triple Quotes: I'm a "Geek" Escaping Single Quote: I'm a "Geek" Escaping Double	Initial String with use of Triple Quotes: Escaped "Easy" Escaping Single Quote: Escaped "Easy" Escaping Double	Initial String with use of Triple Quotes: "Quotes: it's a "Good " Escaping Single Quote: It's a "Good "	Initial String with use of Triple Quotes: I'm a "Concept" Escaping Single Quote: I'm a "Concept"

	Quotes: I'm a "Geek"	Quotes: Escaped "Easy"	Escaping Double Quotes: It's a "Good "	Escaping Double Quotes: I'm a "Concept"
	Escaping Backslashes: C:\Python\Geeks\	Escaping Backslashes: C:\Python\Easy\	Escaping Backslashes: C:\Python\Good\	Escaping Backslashes: C:\Python\Concept\
	Tab: Hi Geeks	Tab: Hi Easy	Tab: Hi Good	Tab: Hi Concept
	New Line: Python Geeks	New Line: Python Easy	New Line: Python Good	New Line: Python Concept

Note:

To ignore the escape sequences in a String, r or R is used, this implies that the string is a raw string and escape sequences inside it are to be ignored.

4. FIND SMALLEST WORD IN SRING

Rohan is given a sequence of words. Can you help him to find the smallest word in that sequence?

Input format

The input test case consists multiple words with different length

Output format

Print the smallest word.

Constraints

Using len() methods

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	Enter any String: Python is simple	Enter any String: Developer are technically strong	Enter any String: You needs more Training	Enter any String: Self-discipline Plays Best roles
Output	Smallest word: is	Smallest word: are	Smallest word: more	Smallest word: Best

5. INDEXING WITH BUILT IN FUNCTION

Create a new string made of the first, middle, and last characters of each input string

Input format

Get the first character from both strings, concatenate them, and store them in variable x

Get the middle character from both strings, concatenate them, and store them in variable y

Output format

Print the result after join x, y, and z

Constraints

Using concatenate operation

Sample test Cases

Test Cases	Test Case 1:	Test Case 2:	Test Case 3:	Test Case 4:
Input	s1 = "America" s2 = "Japan"	s1 = "Korea" s2 = "Japan"	s1 = "America" s2 = "China"	s1 = "Canada" s2 = "Sudan"
Output	AJrpan	KJrpan	ACriaa	CSadan